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After 20 years, the value of the investment is 100

$$\left(1+\frac{r}{100}\right)^{20}=100\left(\left(1+\frac{r}{100}\right)^{5}\right)^{4}=100(1.3001)^{4}=219.75(1.3001)=285.70.$$

After 25 years, the value of the investment is 100

$$\left(1+\frac{r}{100}\right)^{25}=100\left(\left(1+\frac{r}{100}\right)^{5}\right)^{5}=100(1.3001)^{5}=285.70(1.3001)=371.44.$$

After 30 years, the value of the investment is 100

$$\left(1+\frac{r}{100}\right)^{30}=100\left(\left(1+\frac{r}{100}\right)^{5}\right)^{6}=100(1.3001)^{6}=371.44(1.3001)=482.90.$$

***First error*:**

The value of the investment after 10 years was $169.02, but the financial adviser gave $160.02. This was the financial adviser's first error.

The correct answer is $*160.02*.

***Second error*:**

The value of the investment after 25 years was $371.43, but the financial adviser gave $317.43. This was the financial adviser's second error.

The correct answer is $*317.43*.

•

A clothing retailer used to sell only "fast-fashion" pieces, which were low priced and had a profit markup of 50 percent of the per-item cost (including, e.g., the costs of wholesale purchase and marketing). On average, each customer spent $850 annually on around 65 such pieces from the retailer. Now the retailer wishes to double its total profits by selling only "classic" pieces. It plans to double its percentage profit markup per item and generate more revenue per customer while leaving unchanged the company's total costs. The plan assumes that for each classic piece, on average, customers will pay five times what they paid for each fast-fashion piece, and that the total number of customers for the retailer's clothing products will remain the same.

Statements: Customers paid an average of ___1___ dollars (rounded to the nearest dollar) for each of the retailer's fast-fashion pieces. The retailer will need to sell an average minimum of ___2___ classic pieces per person (rounded to the nearest whole number) to achieve its profit goals for classic pieces.

Select values for *1* and for *2* that create the statements that are most strongly supported by the information provided and in accordance with the retailer's plan. Make only two selections, one in each column.

1	**2**	
		10
		13
		15
		17
		21
		25

Answer Explanation:

Strategize

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Strategize

***1*:**

Customers paid an average of $\frac{\$850}{65} \approx \$13.07$ or \$13 rounded to the nearest dollar.

The correct answer is *13*.

***2*:**

For fast-fashion items, let \$C represent the cost per item. For these items, there is a 50% markup on the per-item cost. The customer pays an average of $\frac{\$850}{65}$ per item, so $\frac{\$850}{65} = 1.5(\$C) = \frac{3}{2}(\$C)$. It follows that the retailer's per-item cost \$C is $\frac{2}{3}\left(\frac{\$850}{65}\right)$. The retailer's profit per item is $\frac{\$850}{65} - \frac{2}{3}\left(\frac{\$850}{65}\right) = \frac{1}{3}\left(\frac{\$850}{65}\right)$. The retailer's total profit per customer is $65\left(\frac{1}{3}\left(\frac{\$850}{65}\right)\right) = \frac{\$850}{3}$. For classic items, let \$c be the per-item cost. The retailer plans to double the percentage profit markup of the cost to generate revenue of 2(\$c) per item. However, the customer is willing to pay $5\left(\frac{\$850}{65}\right)$, so the retailer's cost per item, \$c, is $\frac{5}{2}\left(\frac{\$850}{65}\right)$. It follows that the retailer's per-item profit is $5\left(\frac{\$850}{65}\right) - \frac{5}{2}\left(\frac{\$850}{65}\right) = \frac{5}{2}\left(\frac{\$850}{65}\right)$.

The retailer also plans to double the per-customer profit to $\frac{2\,(\$850)}{3}$, and to do this, the retailer will need to sell an average minimum of n items per customer. It follows that $n\left(\frac{5}{2}\left(\frac{\$850}{65}\right)\right) = \frac{2\,(\$850)}{3}$ and

$n = \frac{2\,(\$850)}{3}\,\frac{2\,(65)}{5\,(\$850)} = \frac{52}{3} \approx 17.33$ or 17 rounded to the nearest whole number.

The correct answer is *17*.

Note that the retailer's total cost for fast-fashion items is $\frac{2}{3}\left(\frac{\$850}{65}\right)(65) \approx \$566.67$, and for classic items it is $\frac{5}{2}\left(\frac{\$850}{65}\right)\left(\frac{52}{3}\right) \approx \$566.67$, so the retailer's total costs are left unchanged in the switch from selling fast-fashion items to selling classic items.

•

A certain mail-order company sells T-shirts, buttons, stickers, and current and past issues of its magazine. Each order placed to the company consists of one or more items, and, for each order, the company packages all of the items in the order together. Any employee who packages an order packages the entirety of that order. Among the orders placed to the company today:

None includes both a T-shirt and a magazine.

All of those that include a past issue of the magazine also include the current issue of the magazine.

Most of those that include a magazine also include a sticker.

None of those that include a button also include a sticker.

Statement: For the orders placed to the company today, if one employee were to package all of the orders that include a ___1___, that employee would also package all of the orders that include a ___2___.

Select for *1* and for *2* the two different options that complete the statement so that it is most accurate based on the information provided. Make only two selections, one in each column.

1	2	

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		button
		current issue of the magazine
		past issue of the magazine
		sticker
		T-shirt

Answer Explanation:

Infer

The second statement about the orders placed today implies that all orders containing a past issue of the magazine are among the orders containing a current issue of the magazine. Therefore, if an employee were to package all the orders containing a current issue of the magazine, then the employee would also package all the orders containing a past issue of the magazine.

***1*:**

The correct answer is *current issue of the magazine*.

***2*:**

The correct answer is *past issue of the magazine*.

•

A group of paleobotanists collected samples of sediments containing pollen and spores from various plant species. Each sample was analyzed to determine the number of spores in the sample, the number of pollen grains in the sample, and the species to which those spores and pollen grains belong. Samples S1 and S2 were collected and analyzed in this way. For these two samples:

S1 had the greater total number of spores, and it contained pollen grains from Species P as well as spores from Species Q.

S2 had the greater total number of species represented, and all of the spores it contained were from Species Q.

Statement: Among these two samples, S1 had the greater ratio of total number of ____1____ to total number of ____2____.

Select for *1* and for *2* the options that complete the statement so that it is accurate based on the information provided. Make only two selections, one in each column.

1	2	
		pollen grains
		pollen grains from Species P
		species represented
		spores
		spores from Species Q

Answer Explanation:

Infer

Let (# spores 1), (# spores 2), (# species 1), and (# species 2) be the respective total number of spores in Sample S1, the total number of spores in Sample 2, the total number of species represented in Sample 1, and the total number of

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the total number of spores in Sample 2, the total number of species represented in Sample 1, and the total number of species represented in Sample 2.

(# spores 1) > (# spores 2)	given in first bullet assertion
(# species 1) < (# species 2)	given in second bullet assertion
$\frac{1}{(species\ 1)} > \frac{1}{(species\ 2)}$	take reciprocal of both sides of previous inequality
$\frac{(spores\ 1)}{(species\ 1)} > \frac{(spores\ 2)}{(species\ 2)}$	multiply corresponding left- and right-hand sides of first inequality and previous inequality

The last inequality above states that the ratio of total number of spores to total number of species represented was greater in Sample S1 than in Sample S2.

***1*:**

The correct answer is *spores*.

***2*:**

The correct answer is *species represented*.

Questions 142 to 170 — Difficulty: Hard

142. Biologist: Conservation biologists working to prevent species extinction have long acknowledged that popular species such as lions, eagles, and pandas receive disproportionately large amounts of funding and public attention as compared to less-popular species such as invertebrates and amphibians. Indeed, many of these less-popular species are more in danger of extinction than the more-popular species. Although many conservation biologists have accepted this pattern of disproportionate funding, I believe it needs to stop. For, despite the substantial and continuing expenditure of resources on the more-popular species, very few of these species have any chance of escaping extinction.

The biologist's reasoning is subject to the criticism that the claim that ____A____, which is used to justify the main point, undermines the support for the point that ____B____.

Select for A and for B the options such that criticism of the zoologist's reasoning is strongest. Make only two selections, one in each column.

A	B	
		popular species receive a disproportionate amount of the money and public attention devoted to the preservation of endangered species
		funding and public attention should not be wasted on the preservation of endangered species
		certain popular species are more endangered than many believe
		many species that are not popular are likely to escape extinction
		very few of the more-popular species have any chance of escaping extinction

Answer Explanation:

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Answer Explanation:

Evaluate

***A*:**

The biologist's reasoning starts with the claim that popular species receive disproportionate amounts of funding and public attention compared to less-popular species. The second sentence supports this point by noting that many of the less-popular species are at greater risk of extinction than the more-popular species. The third sentence includes the biologist's main conclusion, that the disproportionate funding of the more-popular species should stop. And the final sentence supports that main conclusion by asserting that very few of the more-popular species have any chance of escaping extinction. But clearly if almost all of the more-popular species have *no chance* of escaping extinction, as the final sentence claims, then the claim in the second sentence that many of the less-popular species are at even *greater* risk of extinction must be false. Thus, the claim in the final sentence, which is used to justify the main point, undermines the claim in the second sentence, which is used to support the point in the first sentence. That is, the claim that very few of the more-popular species have any chance of escaping extinction undermines the claim used to support the point that popular species receive a disproportionate amount of the money and funding devoted to preservation of species.

The correct answer is *very few of the more-popular species have any chance of escaping extinction.*

***B*:**

As explained above, the claim that very few of the more-popular species have any chance of escaping extinction undermines the claim used to support the point that popular species receive a disproportionate amount of the money and funding devoted to preservation of species.

The correct answer is *popular species receive a disproportionate amount of the money and public attention devoted to the preservation of endangered species.*

143. Historian: In the collection of the Science Museum, London, there is a small bottle that is purported to contain the entire original batch of artificial mauve dye created by Sir William Perkin in his laboratory in 1856. Indeed, in his experiments that preceded his commercial production of mauve dyes, he made only a few grams of the substance, but this batch would likely have been completely used up when he tested it as a dye. Also, the early version of mauve dye that Perkin produced commercially consisted of an impure paste rather than the pure crystalline form of the substance contained in the bottle.

Statement: The historian's statements imply that the bottle does not contain ____1____ at least in part because the substance in the bottle is ____2____.

Select for *1* and for *2* the options that create the statement that is most strongly supported by the information above. Make only two selections, one in each column.

1	2	
		any of Perkin's original batch of mauve dye
		a dye produced in Perkin's laboratory
		a dye created in 1856
		a commercially produced dye
		a crystalline form of the dye

Answer Explanation:

Infer

***1*:**

The historian gives two reasons to conclude that the bottle doesn't contain any of Perkin's original batch of mauve dye: first, that the original batch would likely have been completely used up in testing; and second, that the dye in the bottle is crystalline, unlike the earliest version of Perkin's mauve dye produced commercially.

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The correct answer is *any of Perkin's original batch of mauve dye*.

***2*:**

One of the two reasons the historian gives to doubt that the bottle contains any of Perkin's original batch of mauve dye is that the dye in the bottle is crystalline, unlike the earliest version of Perkin's mauve dye produced commercially.

The correct answer is *a crystalline form of the dye*.

144. When a user conducts a search of a certain library database, the database returns the results as a list of items (books, magazines, etc.) that match the search criteria. The user can select from two settings to establish a maximum number of items—either 12 or 24—to be displayed at a time. If the list contains more items than the established maximum, the list will be displayed on multiple pages, with each page (with the possible exception of the last page) containing the established maximum number of items and with each item in the list appearing on exactly one of the pages. The maximum established by the user does not affect the total number of items listed.

For the list of n items returned by a particular search of the database, the number of items, X, displayed on the last page when the established maximum is 12 is different from the number of items, Y, displayed on the last page when the established maximum is 24.

Select for X and for Y values that are jointly consistent with the information provided. Make only two selections, one in each column.

X	Y	
		1
		5
		7
		12
		17

Answer Explanation:

Infer

***X*:**

Since X is the number of the n items left to be displayed on the last page when each previous page displayed exactly 12 items, X must be either 12 or the remainder of $n/12$. Similarly, Y must be either 24 or the remainder of $n/24$. Because 24 is 2 × 12, if $Y \neq X$, then $Y = X + 12$. We are told that $Y \neq X$. Thus, $Y = X + 12$. Among the five response options, only $X = 5$ and $Y = 17$ are such that $Y = X + 12$. That means $X = 5$, so 5 of the n items must be displayed on the last page when the established maximum is 12.

The correct answer is *5*.

***Y*:**

As explained above, $Y = 17$. That is, 17 of the n items must be displayed on the last page when the established maximum is 24.

The correct answer is *17*.

145. In a group of 100 tourists visiting Scandinavia, 10% visited Norway, Sweden, and Denmark; 48% visited only two of these countries; 42% visited only one of these countries; and 100% visited at least one of these countries. Twenty-three tourists in the group visited at least Norway and Sweden, and 28 tourists in the group visited both Norway and Denmark. Furthermore, 3 tourists in the group visited only Denmark, and 32 tourists in the group visited only Sweden.

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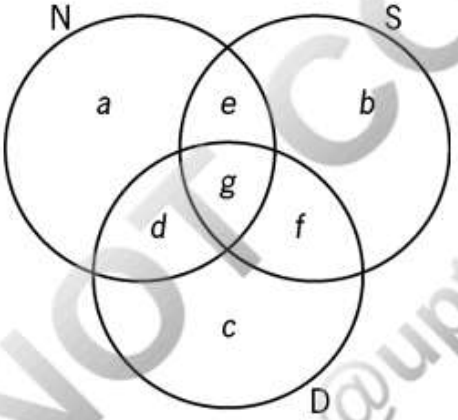
In the table, select the number of tourists in the group who *visited at least Norway*, and select the number of tourists in the group who *visited at least Sweden*. Make only two selections, one in each column.

Visited at least Norway	Visited at least Sweden	
		40
		48
		54
		62
		68
		72

Answer Explanation:

Infer



Consider the Venn diagram above with labels N, S, and D for Norway, Sweden, and Denmark, respectively, and variables a through g representing numbers of the 100 tourists visiting these countries.

Since 10% of the tourists visited Norway, Sweden, and Denmark, it follows that $g = 0.10(100) = 10$.

Also, 28 tourists in the group visited both Norway and Denmark, so $d + g = d + 10 = 28$, from which $d = 18$.

Furthermore, 23 tourists in the group visited at least Norway and Sweden, so $e + g = e + 10 = 23$, from which $e = 13$.

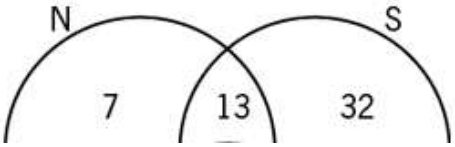
Since 32 tourists in the group visited only Sweden, $b = 32$.

It is given that 48% of the tourists or 0.48(100) = 48 visited only two of the countries, from which it follows that $d + e + f = 48$, so $18 + 13 + f = 48$, from which $f = 17$.

Given that 3 tourists in the group visited only Denmark, $c = 3$.

Given that 42% of the 100 tourists visited only one of these countries, $a + b + c = 0.42(100) = 42$, from which it follows that $a + 32 + 3 = 42$ and $a = 7$.

The Venn diagram below shows the values of a through g.



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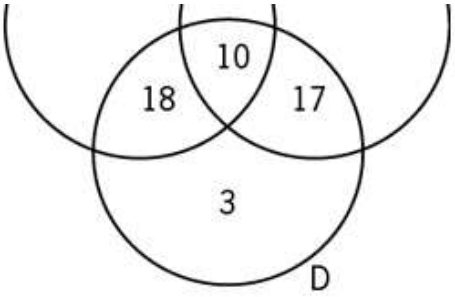

***Visited at least Norway*:**

The number of the 100 tourists who visited at least Norway is 7 + 13 + 10 + 18 = 48.

The correct answer is *48*.

***Visited at least Sweden*:**

The number of the 100 tourists who visited at least Sweden is 13 + 32 + 17 + 10 = 72.

The correct answer is *72*.

146. A city council member standing for reelection has been asked to rate how much she agrees or disagrees that various cultural establishments should receive more city funding. Expressing her overall views regarding budgeting priorities, she answers "Absolutely agree" for the art museum, "Somewhat agree" for the children's theater, "Neither agree nor disagree" for the historical society, "Somewhat disagree" for the symphony orchestra, and "Absolutely disagree" for the metropolitan ballet.

Suppose that the city council member's responses reflect genuine opinions that are logically consistent with her other opinions, and suppose that she believes that funding for any one of the institutions does not affect funding for any of the others. In the first column of the table, rate how much she most clearly agrees that *At least two* of the five institutions should receive more funding. In the second column, rate how much she most clearly agrees that *At most four* of the five institutions should receive more funding. Make only two selections, one in each column.

At least two	**At most four**	
		Absolutely agrees
		Somewhat agrees
		Neither agrees nor disagrees
		Somewhat disagrees
		Absolutely disagrees

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Answer Explanation:

Infer

***At least two*:**

The city council member gave a response of "Absolutely agree" for increased funding for the art museum and "Somewhat agree" for increased funding for the children's theater. Therefore, given the assumptions about her responses, opinions, and beliefs, the city council member clearly somewhat agrees that at least two of the five institutions should receive more funding. That is, there are at least two of the five institutions that it would be false to say that she disagrees or neither agrees nor disagrees with a claim that the institutions should receive increased funding. And, because there is only one institution that she absolutely agrees should receive increased funding, it would be unjustified to conclude that she absolutely agrees that there are two institutions that should receive more funding.

The correct answer is *Somewhat agrees*.

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***At most four*:**

Of the five institutions, there is only one that the city council member gave a response of "Absolutely disagree" for increased funding for, namely the metropolitan ballet. The other four she at most just somewhat disagrees with the idea of increasing funding for. Therefore, given the assumptions about her responses, opinions, and beliefs, the city council member clearly absolutely agrees that at most four of the five institutions should receive more funding. That is, she absolutely does not want all five to have increased funding because she absolutely does not want the metropolitan ballet to receive increased funding.

The correct answer is *Absolutely agrees*.

147. There are exactly three steps—Steps 1, 2, and 3, in that order—used to assemble light bulbs at a particular factory. Some of the bulbs fail to make it through every step in the assembly process; if a bulb fails a given step, it will not proceed to the next step. For the most recent month, the factory manager knows the number of bulbs that were started as well as each of the following:

A = the number of bulbs making it through Step 1 only;

B = the number of bulbs making it through at least Step 2;

C = the number of bulbs making it through all three steps.

The manager wishes to know the fraction of the bulbs that made it through Step 1 that also made it through all three steps. In the table, choose for *Numerator* and *Denominator* the expressions that will give the factory owner the fraction desired. Make only two selections, one in each column.

Numerator	Denominator	
		A
		B
		C
		A + B
		A − C
		B + C

Answer Explanation:

Infer

The fraction of the bulbs that made it through Step 1 (# through Step 1) that also made it through all three steps (# through Steps 1 – 3) is

$$\frac{(\#\text{through Step } 1-3)}{(\#\text{ through Step } 1)} = \frac{(\#\text{through Step } 1-3)}{(\#\text{ through Step 1 then failed}) + (\#\text{ through Step 1 and didn't fail})} =$$

$$\frac{(\#\text{through Step } 1-3)}{(\#\text{ through Step 1 then failed}) + (\#\text{ through at least Step 2})} = \frac{\mathrm{C}}{(\mathrm{A}+\mathrm{B})}$$

***Numerator*:**

In the fraction above, the numerator is C.

The correct answer is *C*.

***Denominator*:**

In the fraction above, the denominator is A + B.

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The correct answer is $A + B$.

148. Legal advocate: The Métis people of Canada are of First Nations and European ancestry. The government grants certain special rights to Métis individuals. To receive these rights, an individual must self-identify as Métis and must not have self-identified only recently in order to receive these rights. The individual must also have Métis ancestry and be accepted as Métis by a modern Métis community. Acceptance by a modern Métis community is shown only by long-standing participation in the community's cultural or political activities.

In the table, select characteristics H and N, such that an individual having H and NOT having N would most clearly satisfy the legal advocate's stated criterion for receiving the rights associated with Métis membership. Make only two selections, one in each column.

H	N	
		Self-identifies as Métis due to long-standing Métis ancestry
		Self-identifies as Métis due to long-standing participation in Métis political activities
		Has recent ancestry that is Métis and participates in Métis political activities
		Has recent ancestry that is not Métis
		Has never participated in Métis political activities

Answer Explanation:

Apply

As a preliminary observation, note that the following three conditions must apply to obtain the special rights:

Self-identify as Métis (not recently)

Have Métis ancestry

Long-standing participation in Métis cultural or political activities

H:

In what follows, possible pairings for H and N will be denoted as

(*Self-identifies as Métis due to long-standing participation in Métis political activities*, not *Has never participated in Métis political activities*),

(*Has recent ancestry that is not Métis*, not *Self-identifies as Métis due to long-standing Métis ancestry*), etc. where, for example, (*Self-identifies as Métis due to long-standing Métis ancestry*, not *Has never participated in Métis political activities*) means the selection of characteristic *Self-identifies as Métis due to long-standing Métis ancestry* for H and characteristic *Has never participated in Métis political activities* for N.

Only characteristics *Has recent ancestry that is not Métis* and *Has never participated in Métis political activities* are such that not having the characteristic could contribute toward receiving the rights, and therefore only characteristics *Has recent ancestry that is not Métis* and *Has never participated in Métis political activities* need to be considered as possibilities for N. Thus, only

(*Self-identifies as Métis due to long-standing Métis ancestry*, not *Has recent ancestry that is not Métis*),

(*Self-identifies as Métis due to long-standing participation in Métis political activities*, not *Has recent ancestry that is not Métis*),

(*Has recent ancestry that is Métis and participates in Métis political activities*, not *Has recent ancestry that is not Métis*),

(*Self-identifies as Métis due to long-standing Métis ancestry*, not *Has never participated in Métis political activities*),

(*Self-identifies as Métis due to long-standing participation in Métis political activities*, not *Has never*

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participated in Métis political activities), and

(*Has recent ancestry that is Métis and participates in Métis political activities*, not *Has never participated in Métis political activities*) need to be considered.

(Note: Below, weakly-(3) means participation in Métis cultural or political activities, but it is not known whether the participation is long-standing.)

(*Self-identifies as Métis due to long-standing Métis ancestry*, not *Has recent ancestry that is not Métis*) — the individual satisfies conditions (1) and (2).

(*Self-identifies as Métis due to long-standing participation in Métis political activities*, not *Has recent ancestry that is not Métis*) — the individual satisfies conditions (1), (2), and (3).

(*Has recent ancestry that is Métis and participates in Métis political activities*, not *Has recent ancestry that is not Métis*) — the individual satisfies conditions (2) and weakly-(3).

(*Self-identifies as Métis due to long-standing Métis ancestry*, not *Has never participated in Métis political activities*) — the individual satisfies conditions (1), (2), and weakly-(3).

(*Self-identifies as Métis due to long-standing participation in Métis political activities*, not *Has never participated in Métis political activities*) — the individual satisfies conditions (1) and (3).

(*Has recent ancestry that is Métis and participates in Métis political activities*, not *Has never participated in Métis political activities*) — the individual satisfies conditions (2) and weakly-(3).

From the above, it follows that the selection of characteristic *Self-identifies as Métis due to long-standing participation in Métis political activities* for H and characteristic *Has recent ancestry that is not Métis* for N most clearly satisfies the legal advocate's criterion.

The correct answer is *Self-identifies as Métis due to long-standing participation in Métis political activities*.

N:

The explanation above shows that the correct selection for N is characteristic *Has recent ancestry that is not Métis*.

The correct answer is *Has recent ancestry that is not Métis*.

149. Tickets for next month's production of *The Sea Gull* at the local community theater went on sale yesterday. The ticket price for a person older than 12 is $\$A$, and the ticket price for a person 12 or under is $\$C$, where $A > C$. Yesterday, the sale of 33 tickets generated a total revenue of \$323, and today the sale of 44 tickets generated a total revenue of \$424.

In the table, select a value for A and a value for C that are jointly consistent with the given information. Make only two selections, one in each column.

A	C	
		6
		8
		10
		11
		12

Answer Explanation:

Infer

Note that 44 − 33 = 11 more tickets were sold today than yesterday and \$424 − \$323 = \$101 more revenue was generated today than yesterday. If a represents the number of these 11 tickets that were adult tickets (i.e., tickets for those older than 12), then $11 - a$ represents the number of these 11 tickets that were child tickets (i.e.,

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for those older than 12), then $11 - a$ represents the number of these 11 tickets that were child tickets (i.e., tickets for those 12 or under) and $aA + (11 - a)C = 101$ represents the revenue, in dollars, generated by the sale of these 11 tickets.

Note that at least one of A or C must be odd, because if both are even, then $aA + (11 - a)C$ is even and cannot be 101. Of the possible values of A and C in the answer choices, 11 is the only odd number. Therefore, $A = 11$ or $C = 11$. If $A = 11$, then the possible choices for C are 6, 8, and 10. Note that $C \neq 12$ because $A > C$. If $C = 11$, then $A = 12$ because $A > C$.

If $A = 11$ and $C = 6$,

$$\begin{aligned} aA + (11 - a)C &= 101 && \text{given} \\ 11a + 6(11 - a) &= 101 && \text{substitution} \\ 11a + 66 - 6a &= 101 && \text{distributive property} \\ 5a + 66 &= 101 && \text{combine like terms} \\ 5a &= 35 && \text{subtract 66 from both sides} \\ a &= 7 && \text{divide both sides by 5} \end{aligned}$$

This is a valid value for a.

If $A = 11$, and $C = 8$, then

$$\begin{aligned} aA + (11 - a)C &= 101 && \text{given} \\ 11a + 8(11 - a) &= 101 && \text{substitution} \\ 11a + 88 - 8a &= 101 && \text{distributive property} \\ 3a + 88 &= 101 && \text{combine like terms} \\ 3a &= 13 && \text{subtract 88 from both sides} \\ a &= 133 && \text{divide both sides by 3} \end{aligned}$$

Since a must be a nonnegative integer, 133 is not a valid value for a.

If $A = 11$ and $C = 10$, then

$$\begin{aligned} aA + (11 - a)C &= 101 && \text{given} \\ 11a + 10(11 - a) &= 101 && \text{substitution} \\ 11a + 110 - 10a &= 101 && \text{distributive property} \\ a + 110 &= 101 && \text{combine like terms} \\ a &= -9 && \text{subtract 110 from both sides} \end{aligned}$$

Since a must be a nonnegative integer, −9 is not a valid value for a.

If $A = 12$ and $C = 11$, then

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$$aA + (11 - a)C = 101 \quad \text{given}$$

$$12a + 11(11 - a) = 101 \quad \text{substitution}$$

$$12a + 121 - 11a = 101 \quad \text{distributive property}$$

$$a + 121 = 101 \quad \text{combine like terms}$$

$$a = -20 \quad \text{subtract 121 from both sides}$$

Since a must be a nonnegative integer, −20 is not a valid value for a.

***A*:**

From the work above, letting $A = 11$ and $C = 6$ gives the only valid value for a.

The correct answer is *11*.

***C*:**

From the work above, letting $A = 11$ and $C = 6$ gives the only valid value for a.

The correct answer is *6*.

Note that $A = 11$ and $C = 6$ are consistent with the given information. First, since 11 > 6, $A > C$. Also, there exist numbers of adult and child tickets sold yesterday, namely 25 and 8, respectively, such that 33 tickets (25 + 8 = 33) were sold yesterday and they generated revenue of (25)(11) + (8)(6) = 275 + 48 = 323 dollars. Furthermore, there exist numbers of adult and child tickets sold today, namely 32 and 12, respectively, such that 44 tickets (32 + 12 = 44) were sold yesterday and they generated revenue of (32)(11) + (12)(6) = 352 + 72 = 424 dollars.

150. Companies A and B are part of the same industry and are located in the same city. For Company A, the average (arithmetic mean) salary of its employees, in United Arab Emirates dirhams (AED), is 10,000 AED higher than that for Company B. However, more than half of the employees at Company A have salaries below the average for Company B.

Statement: If the average salary at Company B is ___1___, then the median salary at Company A is ___2___.

Select for *1* and for *2* the options that complete the statement so that it most accurately reflects the information provided. Make only two selections, one in each column.

1	2	
		greater than 100,000 AED
		less than 100,000 AED
		equal to 110,000 AED
		between 100,000 and 110,000 AED
		greater than 110,000 AED

Answer Explanation:

Infer

The statement that more than half of the employees at Company A have salaries below the average salary at Company B is equivalent to the median salary at Company A is less than the average salary at Company B.

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